PYTHON PROGRAMMING

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Class:II BCA

LABCYCLE-1

1.Write a program to count vowels, consonants, digits, and specialcharacters in a string given as input.

PROGRAM:

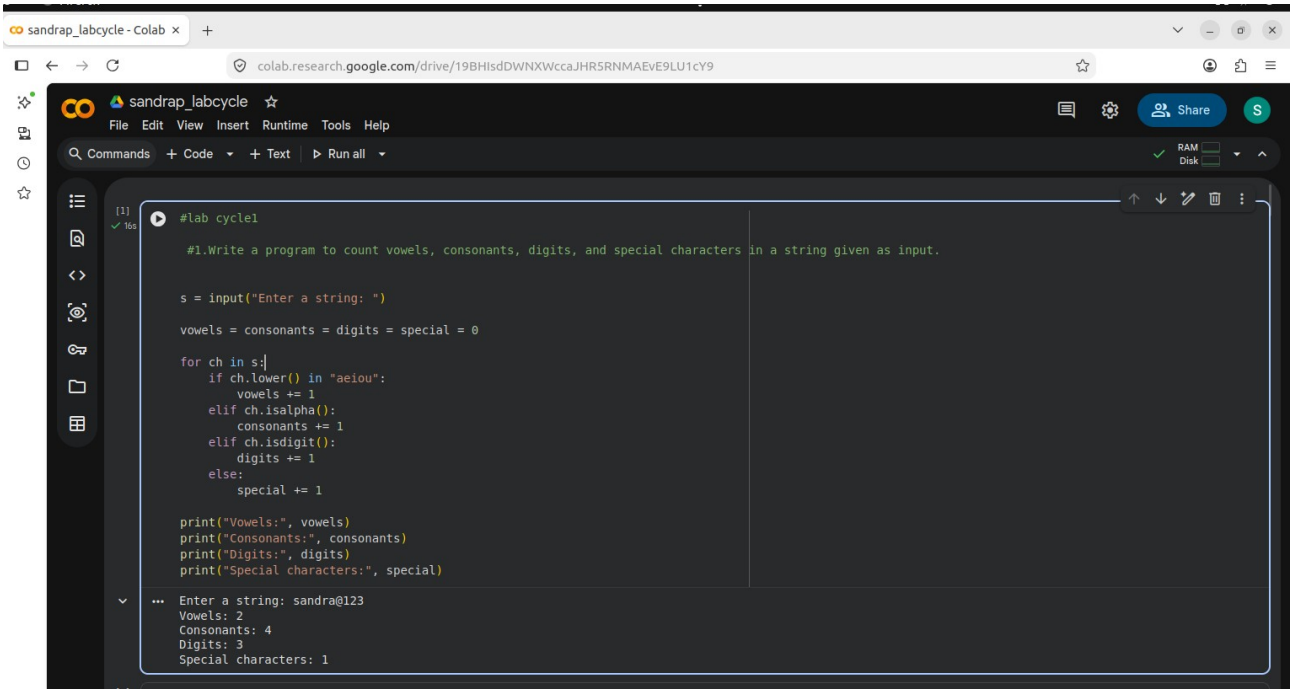
```
 s = input("Enter a string: ")

vowels = consonants = digits = special = 0

for ch in s:
    if ch.lower() in "aeiou":
        vowels += 1
    elif ch.isalpha():
        consonants += 1
    elif ch.isdigit():
        digits += 1
    else:
        special += 1

print("Vowels:", vowels)
print("Consonants:", consonants)
print("Digits:", digits)
print("Special characters:", special)
```

OUTPUT:



2. Write a program to find the largest and second largest element in a list.

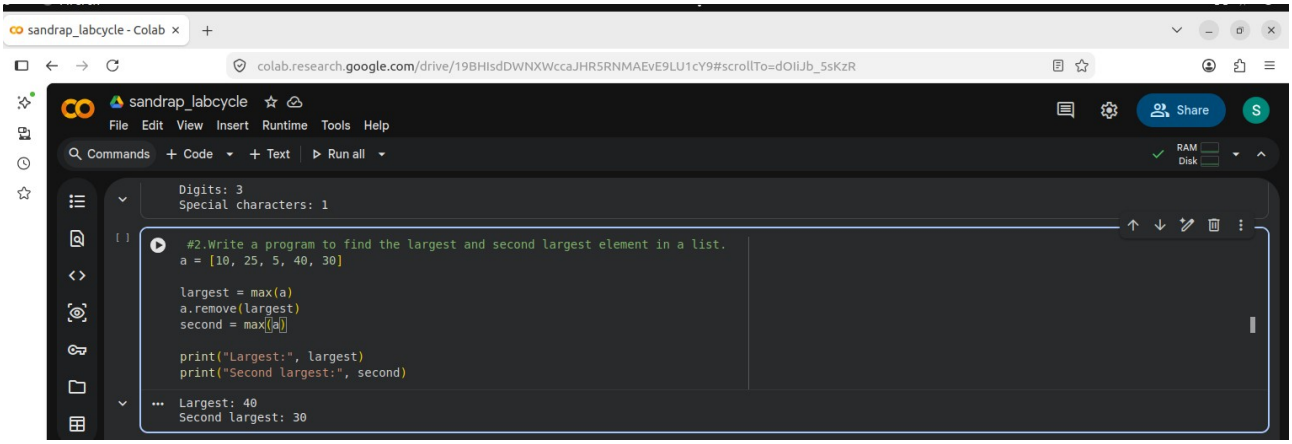
PROGRAM:

```
a = [10, 25, 5, 40, 30]

largest = max(a)
a.remove(largest)
second = max(a)

print("Largest:", largest)
print("Second largest:", second)
```

OUTPUT:



3.Write a program to count word frequency using a dictionary.

PROGRAM:

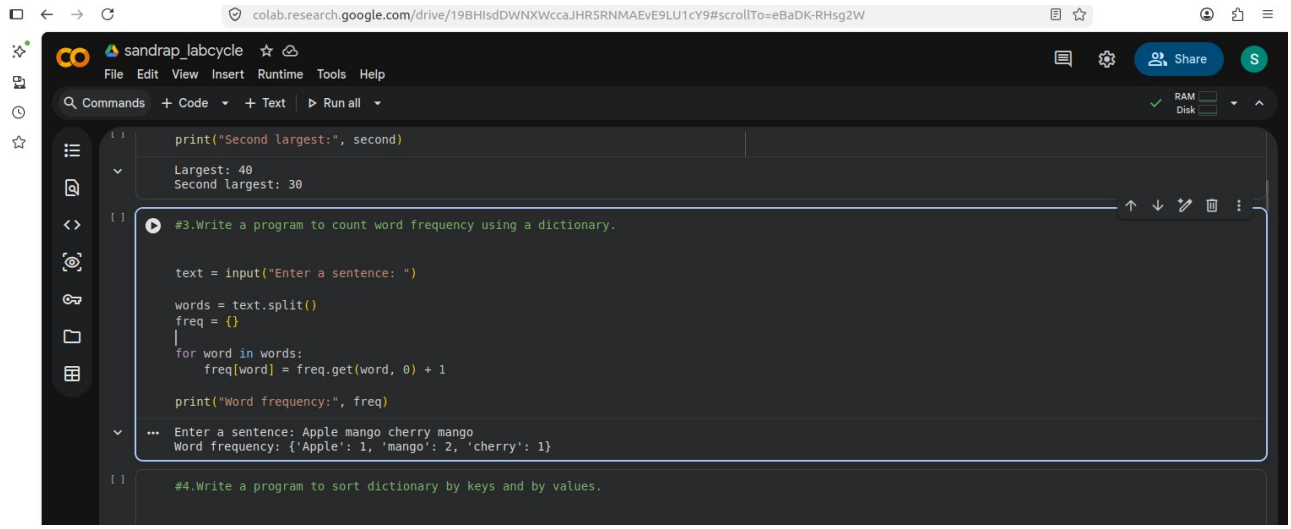
```
text = input("Enter a sentence: ")

words = text.split()
freq = {}

for word in words:
    freq[word] = freq.get(word, 0) + 1

print("Word frequency:", freq)
```

OUTPUT:

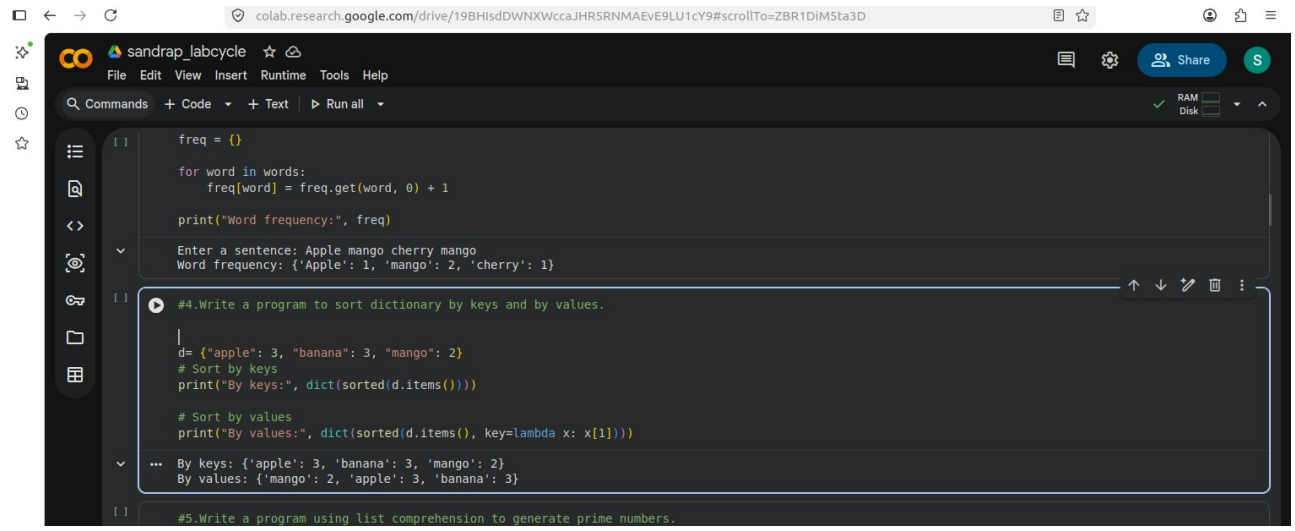


4.Write a program to sort dictionary by keys and the by values

PROGRAM:

```
d={'b': 20, 'a': 10, 'c': 30}
print("By keys:", dict(sorted(d.items())))
for k, v in sorted(d.items()):
    print(k, v)
```

OUTPUT:

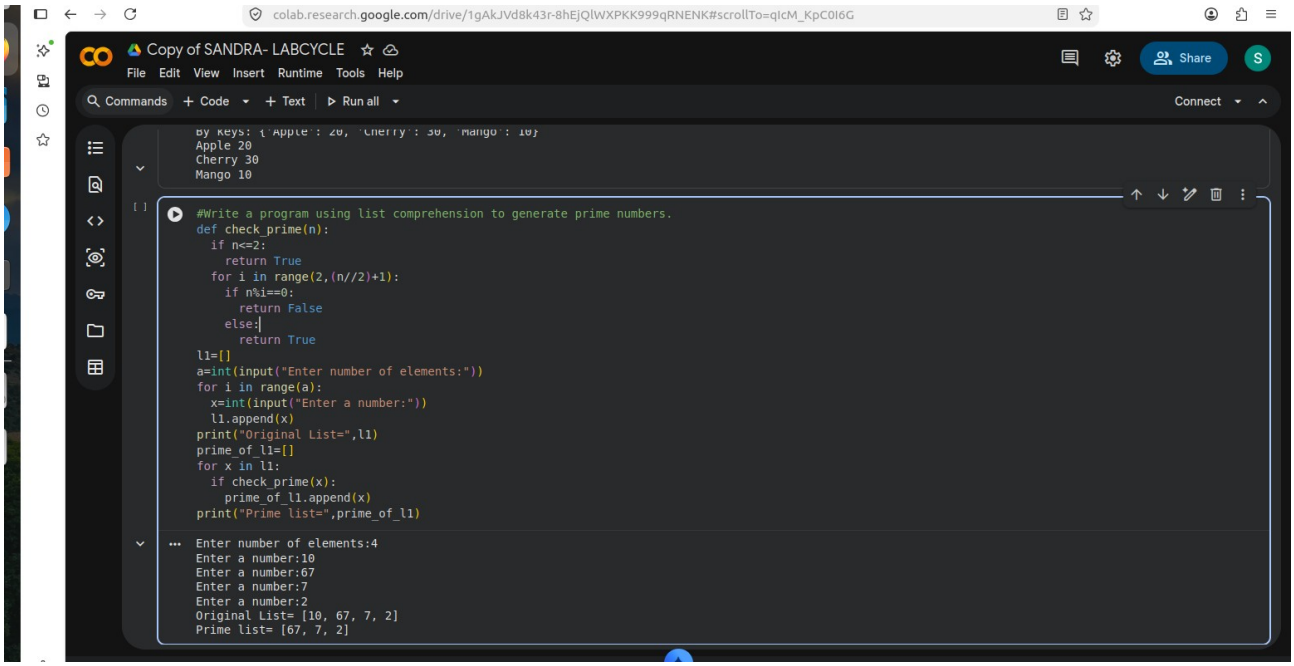


5.Write a program using list comprehension to generate prime numbers.

PROGRAM:

```
def check_prime(n):
  if n<=2:
    return True
  for i in range(2,(n//2)+1):
    if n%i==0:
      return False
    else:
      return True
l1=[]
a=int(input("Enter number of elements:"))
for i in range(a):
  x=int(input("Enter a number:"))
  l1.append(x)
print("Original List=",l1)
prime_of_l1=[]
for x in l1:
  if check_prime(x):
    prime_of_l1.append(x)
print("Prime list=",prime_of_l1)
```

OUTPUT:



LABCYCLE-2

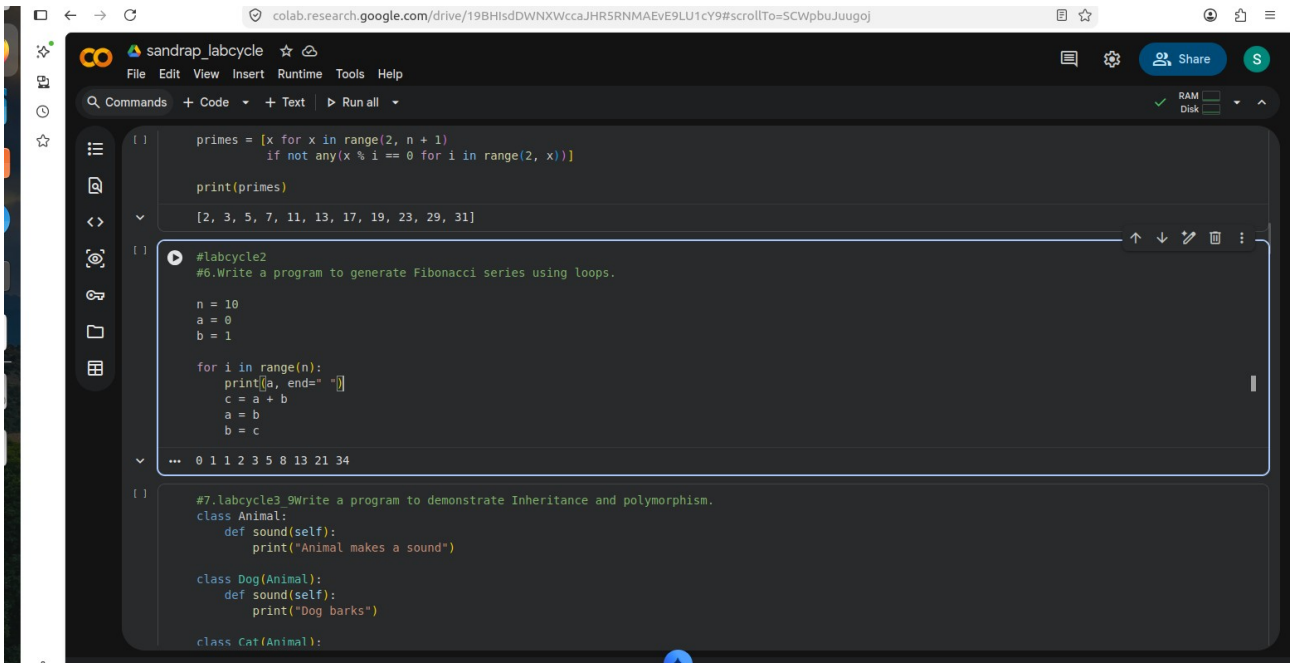
6.Write a program to generate Fibonacci series using loops.

PROGRAM:

```
n = 10
a = 0
b = 1

for i in range(n):
    print(a, end=" ")
    c = a + b
    a = b
    b = c
```

OUTPUT:

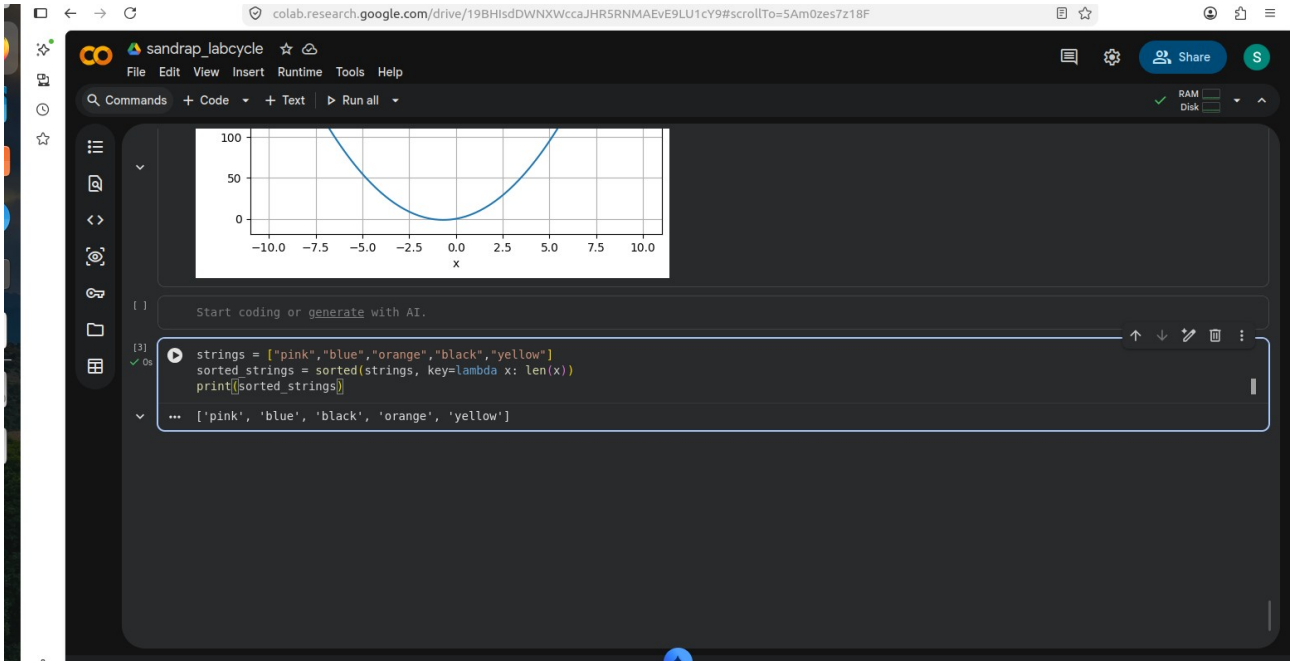


7.Sort strings by length using lambda function.

PROGRAM:

```
strings = ["pink","blue","orange","black","yellow"]
sorted_strings = sorted(strings, key=lambda x: len(x))
print(sorted_strings)
```

OUTPUT:

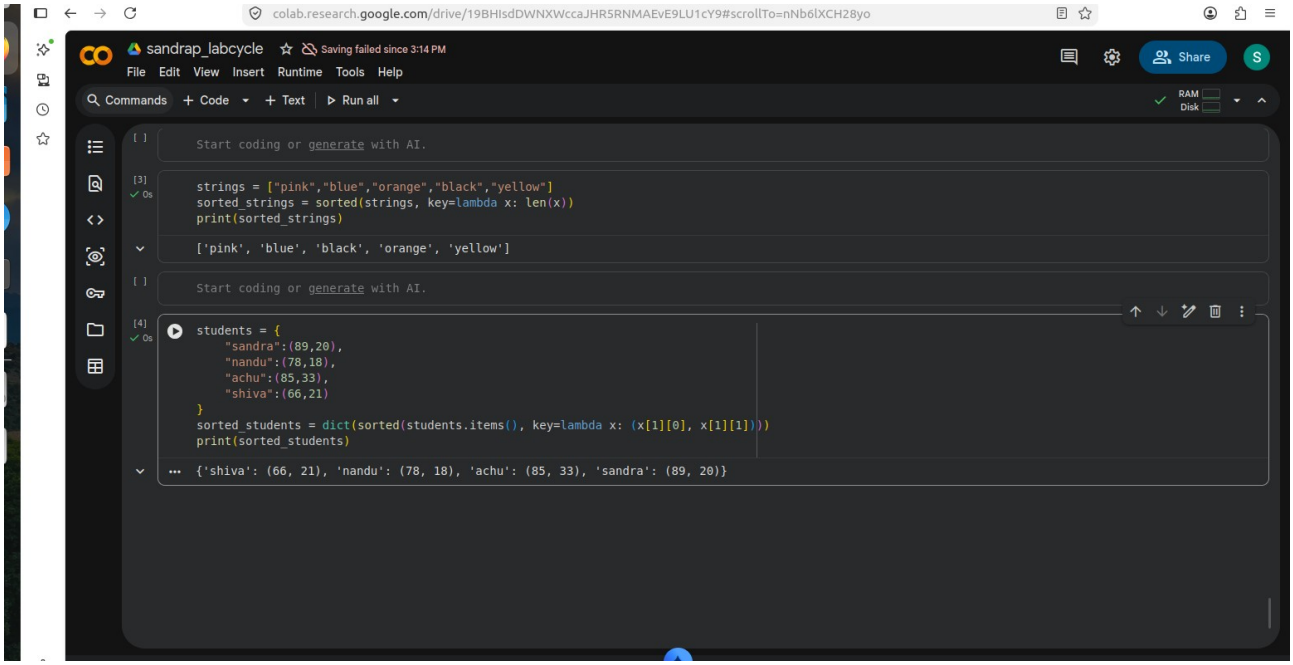


8.Sort dictionary using multiple values and lambda function.

PROGRAM:

```
students = {
    "sandra":(89,20),
    "nandu":(78,18),
    "achu":(85,33),
    "shiva":(66,21)
}
sorted_students = dict(sorted(students.items(), key=lambda x: (x[1][0], x[1][1])))
print(sorted_students)
```

OUTPUT:



LABCYCLE-3

9.Write a program to demonstrate inheritance and polymorphism.

PROGRAM:

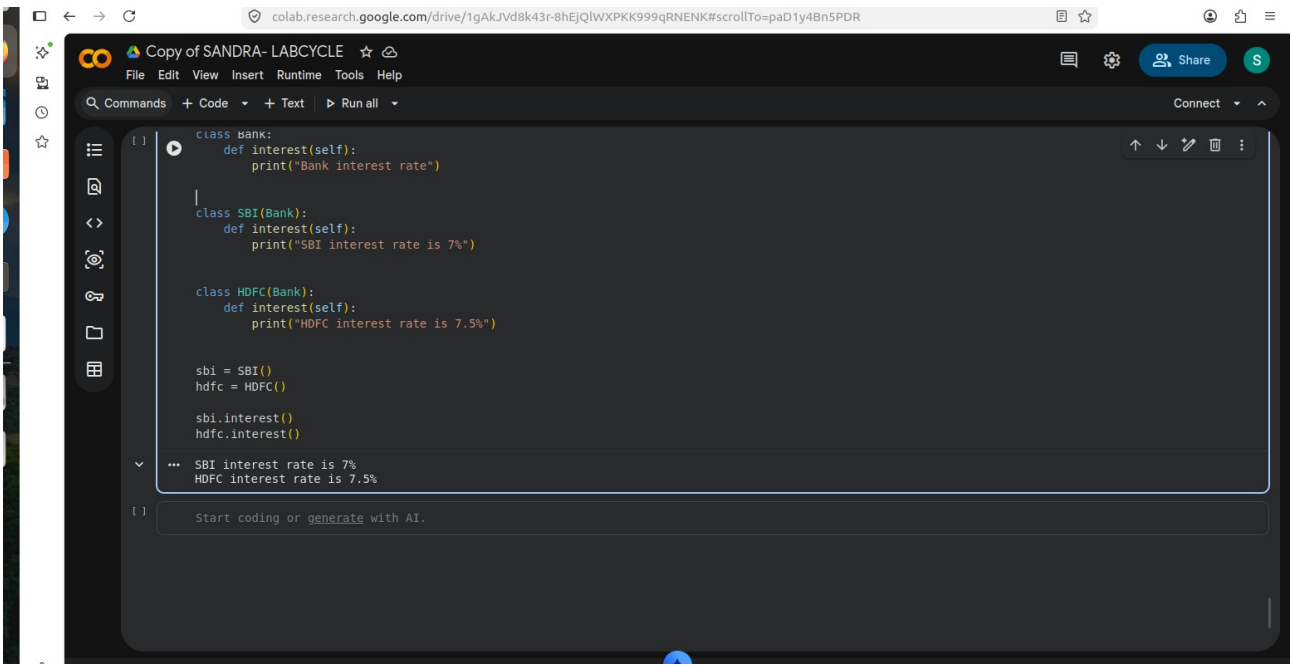
```
class Bank:
    def interest(self):
        print("Bank interest rate")

class SBI(Bank):
    def interest(self):
        print("SBI interest rate is 7%")

class HDFC(Bank):
    def interest(self):
        print("HDFC interest rate is 7.5%")

sbi = SBI()
hdfc = HDFC()
sbi.interest()
hdfc.interest()
```

OUTPUT:



10.Write a program to read and write text files .Use exceptions

PROGRAM:

```
try:
    with open("example.txt", "w") as file:
        text = input("Enter text to write to the file: ")
        file.write(text)

    with open("example.txt", "r") as file:
        content = file.read()

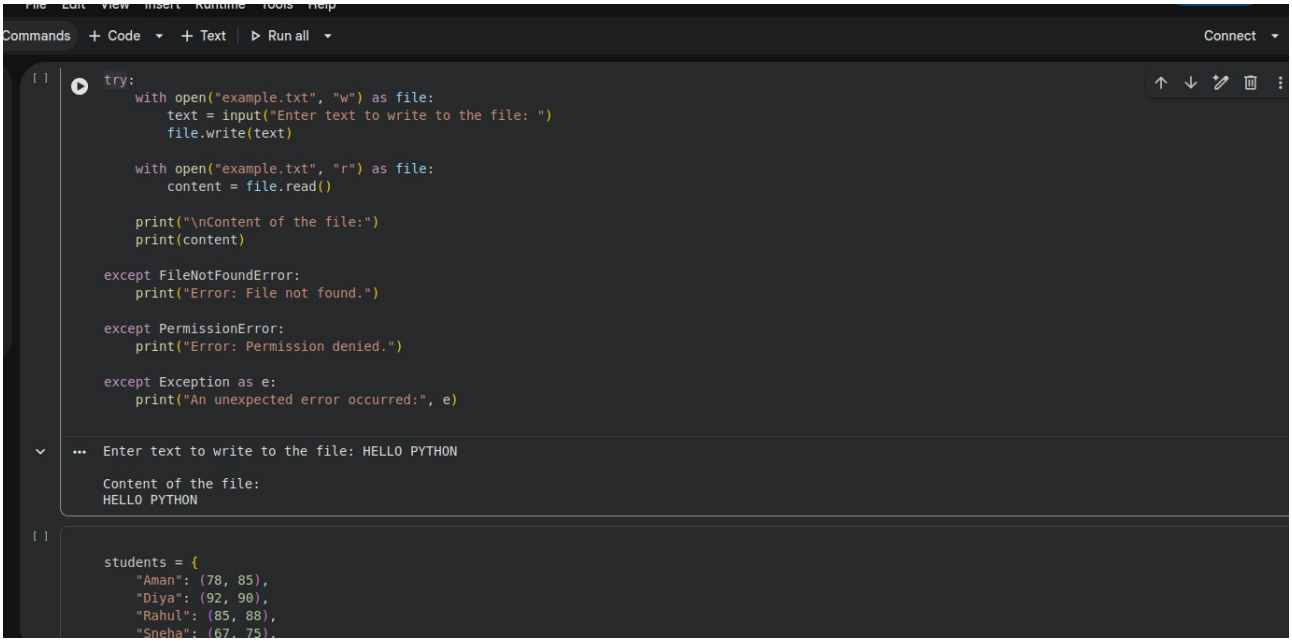
    print("\nContent of the file:")
    print(content)

except FileNotFoundError:
    print("Error: File not found.")

except PermissionError:
    print("Error: Permission denied.")

except Exception as e:
    print("An unexpected error occurred:", e)
```

OUTPUT:



LABCYCLE-4

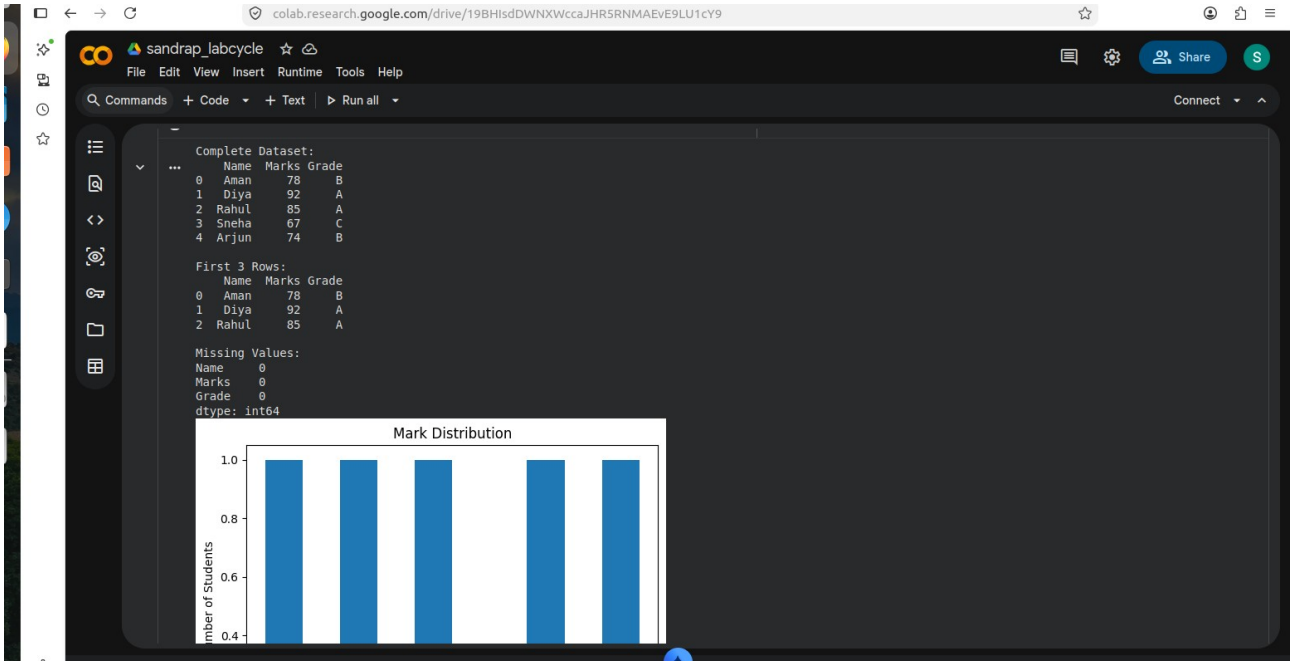
11.Write a Python program using Pandas
a. to create and analyze a student dataset containing student name, marks and grade.
b. Use the following data: data = { "Name": ["Aman", "Diya", "Rahul", "Sneha",
"Arjun"], "Marks": [78, 92, 85, 67, 74], "Grade": ["B", "A", "A", "C", "B"]}
c. Perform the following operations:
i. Create a Pandas DataFrame using the given dataset.
ii. Display the complete dataset.
iii. Display the first 3 rows of the dataset.
iv. Check whether the dataset contains missing values.
v. Create a histogram showing mark distribution.
vi. Find median and standard deviation of marks.

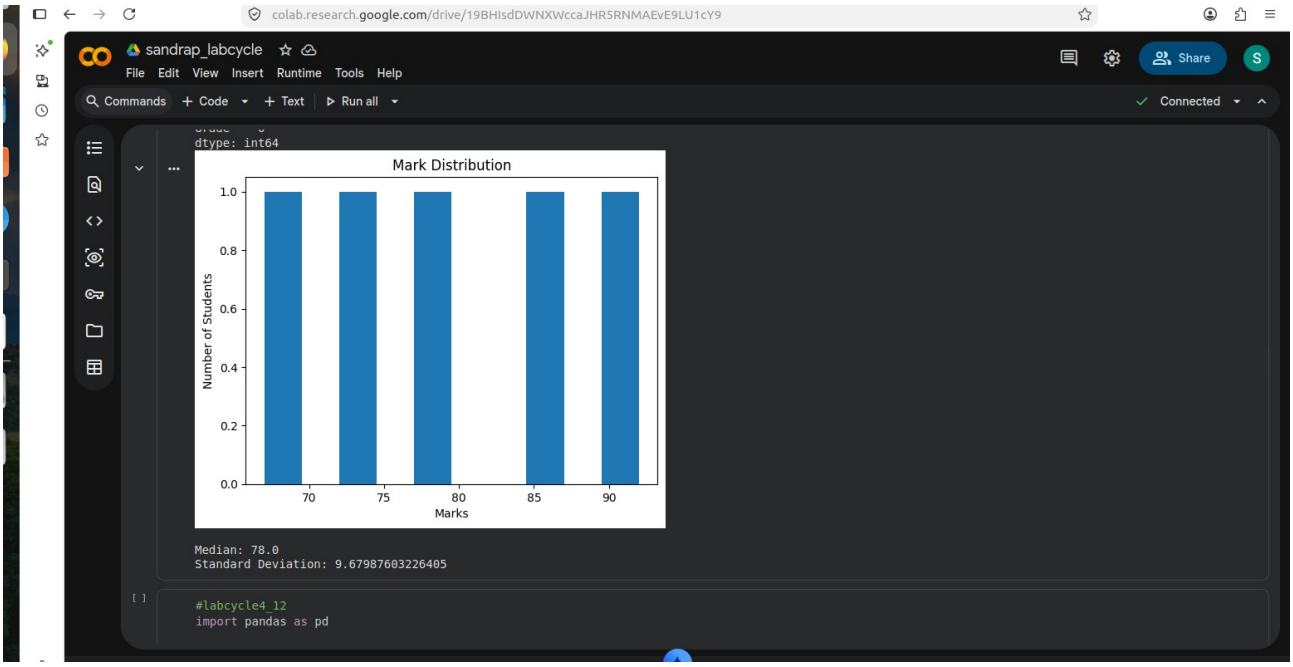
PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Name": ["Aman", "Diya", "Rahul", "Sneha", "Arjun"],
    "Marks": [78, 92, 85, 67, 74],
    "Grade": ["B", "A", "A", "C", "B"]
}
df = pd.DataFrame(data)
print("Complete Dataset:")
print(df)
print("\nFirst 3 Rows:")
print(df.head(3))
print("\nMissing Values:")
print(df.isnull().sum())
plt.hist(df["Marks"])
plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.title("Mark Distribution")
plt.show()
print("\nMedian:", df["Marks"].median())
print("Standard Deviation:", df["Marks"].std())
```

OUTPUT:


colab.research.google.com/drive/19BHIsdDWNXWccaJHR5RNMAEvE9LU1cY9
sandrap_labcycle
File Edit View Insert Runtime Tools Help
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Connect
Complete Dataset:
Name Marks Grade
0 Aman 78 B
1 Diya 92 A
2 Rahul 85 A
3 Sneha 67 C
4 Arjun 74 B
First 3 Rows:
Name Marks Grade
0 Aman 78 B
1 Diya 92 A
2 Rahul 85 A
Missing Values:
Name 0
Marks 0
Grade 0
dtype: int64
Mark Distribution
1.0
0.8
0.6
0.4
mber of Students


colab.research.google.com/drive/19BHIsdDWNXWccaJHR5RNMAEvE9LU1cY9
sandrap_labcycle
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Connected
dtype: int64
Mark Distribution
1.0
0.8
0.6
0.4
0.2
0.0
Number of Students
70
75
80
85
90
Marks
Median: 78.0
Standard Deviation: 9.67987603226405
#labcycle4_12
import pandas as pd

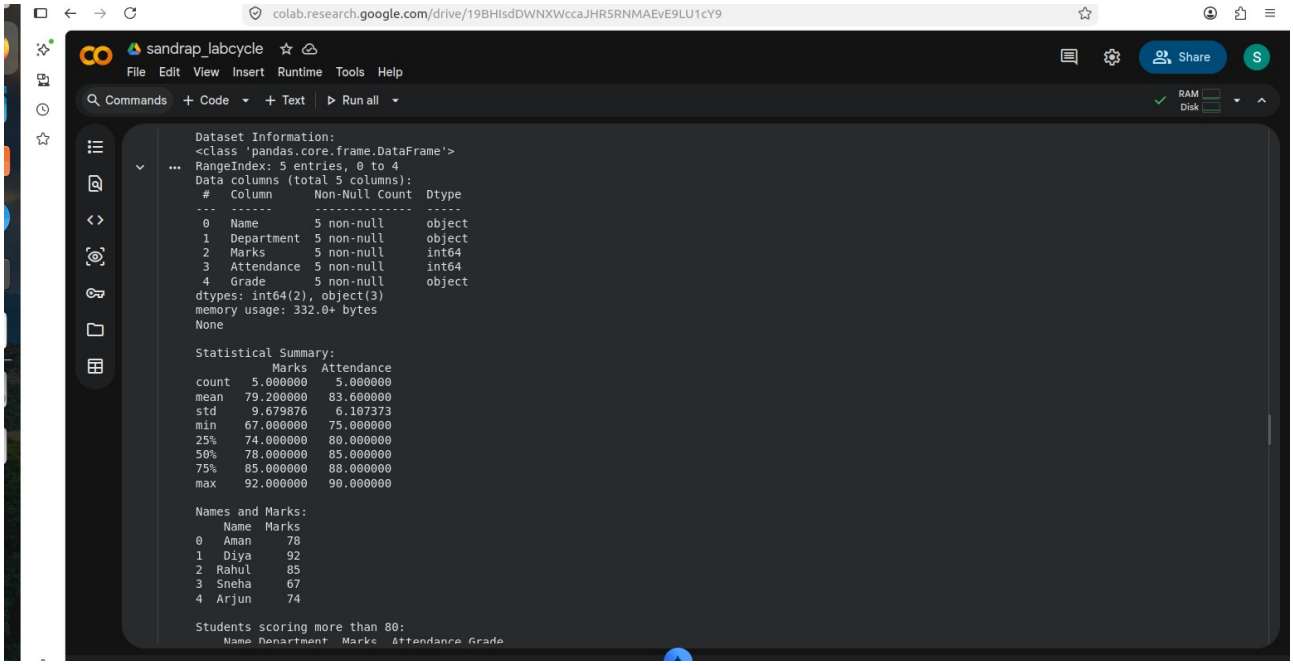
12.Create dataframe from the given
a. data = {"Name": ["Aman", "Diya", "Rahul", "Sneha", Arjun"], "Department":
["CS", "IT", "CS", "ECE", "IT"], "Marks": [78, 92, 85, 67, 74], "Attendance":
[85, 90, 88, 75, 80], "Grade": ["B", "A", "A", "C", "B"]}.
b. Display Dataset Information
c. Display Statistical Summary
d. Display only student names and marks
e. Display students scoring more than 80 marks.
f. Sort students based on marks in descending order.
g. Add a new column named “Bonus” containing: 5 marks bonus for every student.

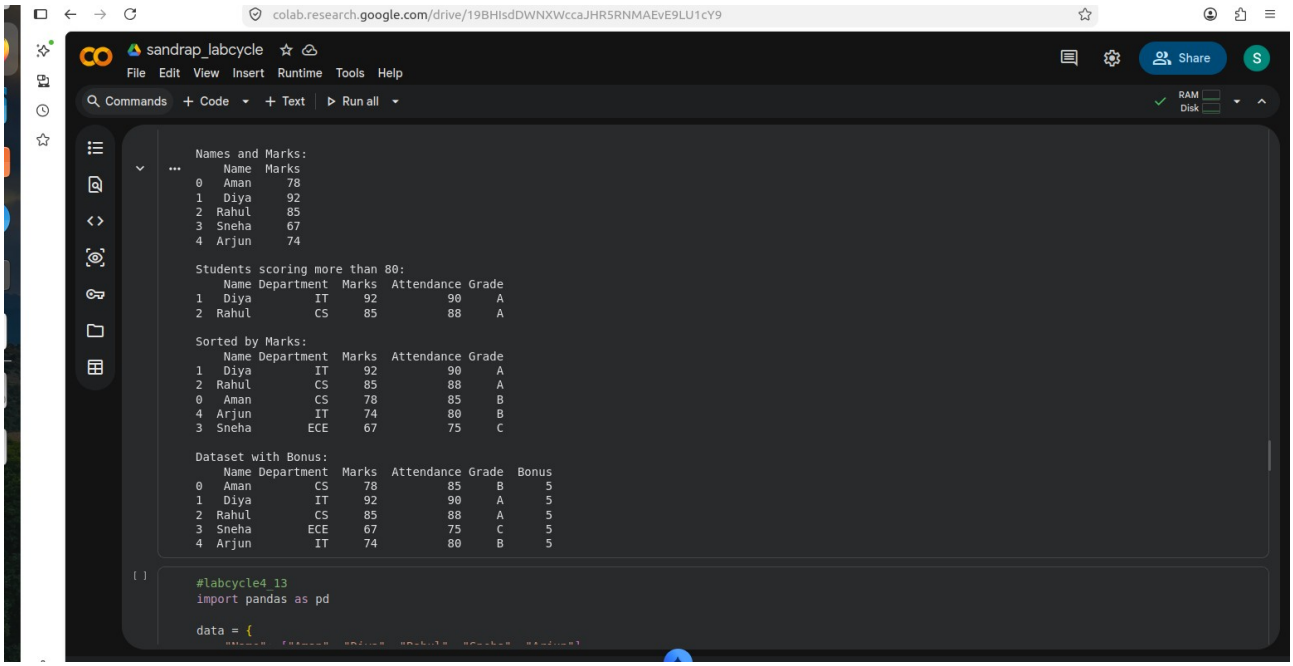
PROGRAM:

```
import pandas as pd
data = {
    "Name": ["Aman", "Diya", "Rahul", "Sneha", "Arjun"],
    "Department": ["CS", "IT", "CS", "ECE", "IT"],
    "Marks": [78, 92, 85, 67, 74],
    "Attendance": [85, 90, 88, 75, 80],
    "Grade": ["B", "A", "A", "C", "B"]
}

df = pd.DataFrame(data)
print("Dataset Information:")
print(df.info())
print("\nStatistical Summary:")
print(df.describe())
print("\nNames and Marks:")
print(df[["Name", "Marks"]])
print("\nStudents scoring more than 80:")
print(df[df["Marks"] > 80])
print("\nSorted by Marks:")
print(df.sort_values("Marks", ascending=False))
df["Bonus"] = 5
print("\nDataset with Bonus:")
print(df)
```

OUTPUT:


colab.research.google.com/drive/19BHIsdDWNXWccaJHR5RNMAEvE9LU1cY9
sandrap_labcycle
File Edit View Insert Runtime Tools Help
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RAM
Disk
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 5 columns):
Column Non-Null Count Dtype
0 Name 5 non-null object
1 Department 5 non-null object
2 Marks 5 non-null int64
3 Attendance 5 non-null int64
4 Grade 5 non-null object
dtypes: int64(2), object(3)
memory usage: 332.0+ bytes
None
Statistical Summary:
Marks Attendance
count 5.000000 5.000000
mean 79.200000 83.600000
std 9.679876 6.107373
min 67.000000 75.000000
25% 74.000000 80.000000
50% 78.000000 85.000000
75% 85.000000 88.000000
max 92.000000 90.000000
Names and Marks:
Name Marks
0 Aman 78
1 Diya 92
2 Rahul 85
3 Sneha 67
4 Arjun 74
Students scoring more than 80:


colab.research.google.com/drive/19BHIsdDWNXWccaJHR5RNMAEvE9LU1cY9
sandrap_labcycle
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM
Disk
Names and Marks:
Name Marks
0 Aman 78
1 Diya 92
2 Rahul 85
3 Sneha 67
4 Arjun 74
Students scoring more than 80:
Name Department Marks Attendance Grade
1 Diya IT 92 90 A
2 Rahul CS 85 88 A
Sorted by Marks:
Name Department Marks Attendance Grade
1 Diya IT 92 90 A
2 Rahul CS 85 88 A
0 Aman CS 78 85 B
4 Arjun IT 74 80 B
3 Sneha ECE 67 75 C
Dataset with Bonus:
Name Department Marks Attendance Grade Bonus
0 Aman CS 78 85 B 5
1 Diya IT 92 90 A 5
2 Rahul CS 85 88 A 5
3 Sneha ECE 67 75 C 5
4 Arjun IT 74 80 B 5
#labcycle4_13
import pandas as pd
data = {

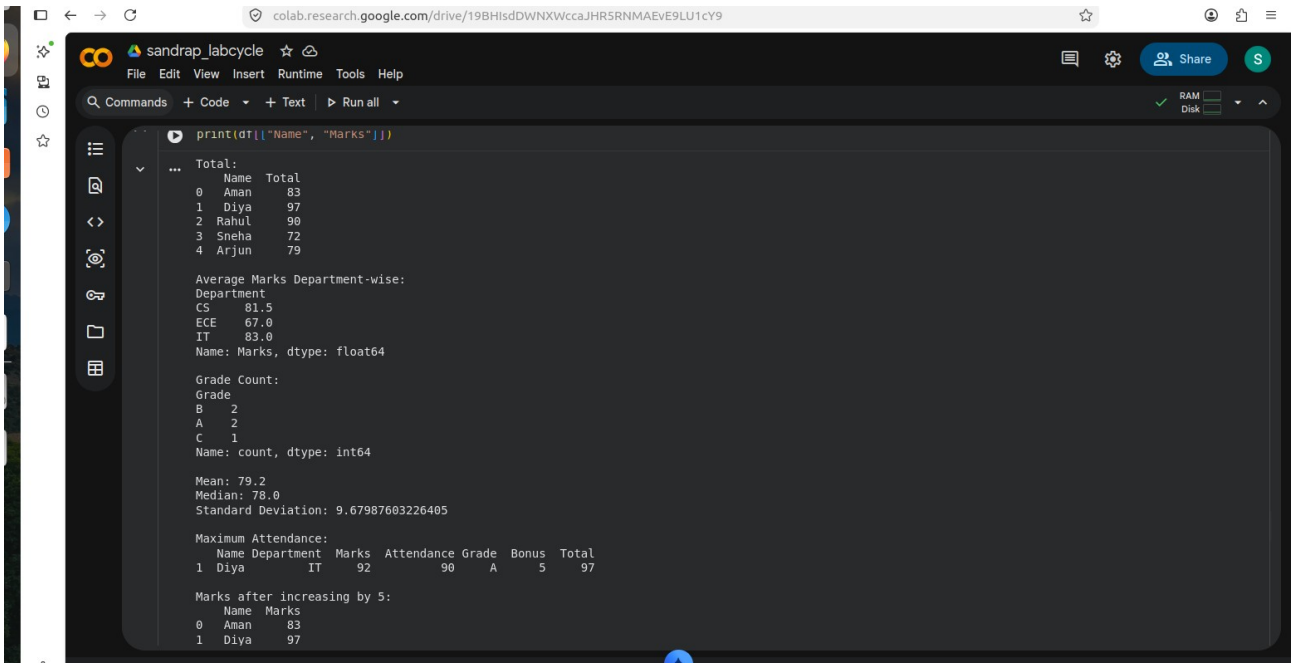
13.Using the above dataframe answer the following questions.
a. Create a new column named “Total” by adding marks and bonus.
b. Find average marks department-wise.
c. Count the number of students in each grade category.
d. Calculate: mean, median,standard deviation of student marks.
e. Display students having maximum attendance.
f. Increase marks of all students by 5.

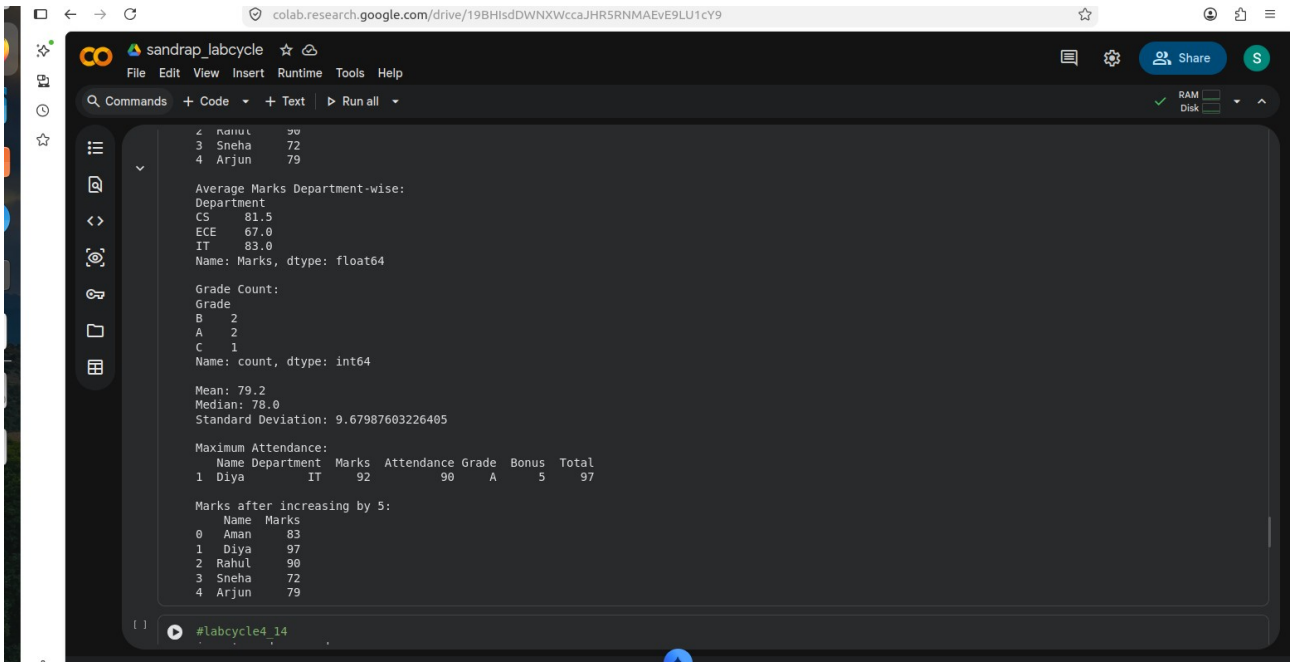
PROGRAM:

```
import pandas as pd
data = {
    "Student": ["A", "B", "C", "D", "E", "F", "G", "H"],
    "Department": ["CSE", "ECE", "CSE", "EEE", "ECE", "CSE", "EEE", "ECE"],
    "Marks": [75, 82, 68, 90, 76, 88, 65, 79],
    "Bonus": [5, 3, 7, 2, 4, 5, 6, 3],
    "Grade": ["B", "A", "C", "A", "B", "A", "C", "B"],
    "Attendance": [85, 92, 78, 95, 88, 95, 75, 90]
}
df = pd.DataFrame(data)
print("Original DataFrame:")
print(df)df["Total"] = df["Marks"] + df["Bonus"]
print("\nDataFrame after adding Total:")
print(df)
average_marks = df.groupby("Department")["Marks"].mean()
print("\nAverage Marks Department-wise:")
print(average_marks)
grade_count = df["Grade"].value_counts()
print("\nNumber of Students in Each Grade:")
print(grade_count)
mean_marks = df["Marks"].mean()
median_marks = df["Marks"].median()
standard_deviation = df["Marks"].std()
print("\nStatistics of Student Marks:")
print("Mean =", mean_marks)
print("Median =", median_marks)
print("Standard Deviation =", standard_deviation)
maximum_attendance = df["Attendance"].max()
print("\nMaximum Attendance =", maximum_attendance)
students_max_attendance = df[
    df["Attendance"] == maximum_attendance
]
print("\nStudents Having Maximum Attendance:")
print(students_max_attendance)
df["Marks"] = df["Marks"] + 5

print("\nDataFrame after Increasing Marks by 5:")
print(df)
```

OUTPUT:


colab.research.google.com/drive/19BHIsdDWNXWccaJHR5RNMAEvE9LU1cY9
sandrap_labcycle
File Edit View Insert Runtime Tools Help
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RAM Disk
print(df[["Name", "Marks"]])
Total:
Name Total
0 Aman 83
1 Diya 97
2 Rahul 90
3 Sneha 72
4 Arjun 79
Average Marks Department-wise:
Department
CS 81.5
ECE 67.0
IT 83.0
Name: Marks, dtype: float64
Grade Count:
Grade
B 2
A 2
C 1
Name: count, dtype: int64
Mean: 79.2
Median: 78.0
Standard Deviation: 9.67987603226405
Maximum Attendance:
Name Department Marks Attendance Grade Bonus Total
1 Diya IT 92 90 A 5 97
Marks after increasing by 5:
Name Marks
0 Aman 83
1 Diya 97


colab.research.google.com/drive/19BHIsdDWNXWccaJHR5RNMAEvE9LU1cY9
sandrap_labcycle
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM Disk
3 Sneha 72
4 Arjun 79
Average Marks Department-wise:
Department
CS 81.5
ECE 67.0
IT 83.0
Name: Marks, dtype: float64
Grade Count:
Grade
B 2
A 2
C 1
Name: count, dtype: int64
Mean: 79.2
Median: 78.0
Standard Deviation: 9.67987603226405
Maximum Attendance:
Name Department Marks Attendance Grade Bonus Total
1 Diya IT 92 90 A 5 97
Marks after increasing by 5:
Name Marks
0 Aman 83
1 Diya 97
2 Rahul 90
3 Sneha 72
4 Arjun 79
#labcycle4_14

LAB CYCLE-5

15.Plot simple mathematics functions.[y=3x^2+4x]

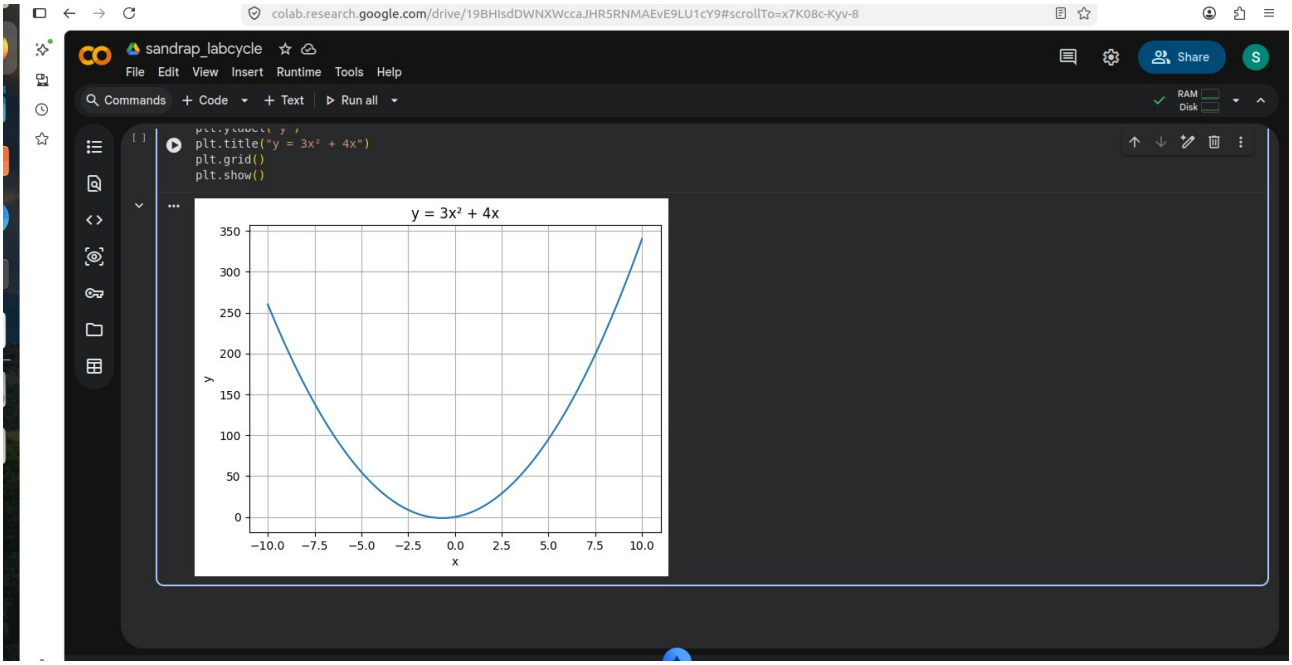
PROGRAM:

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-10, 10, 100)
y = 3*x**2 + 4*x

plt.plot(x, y)
plt.xlabel("x")
plt.ylabel("y")
plt.title("y = 3x² + 4x")
plt.grid()
plt.show()
```

OUTPUT:



16. Create simple login form using streamlit.

PROGRAM:

```
%%writefile sa-gui.py

import streamlit as st
st.title("Login")
username = st.text_input("Username")
password = st.text_input("Password", type="password")

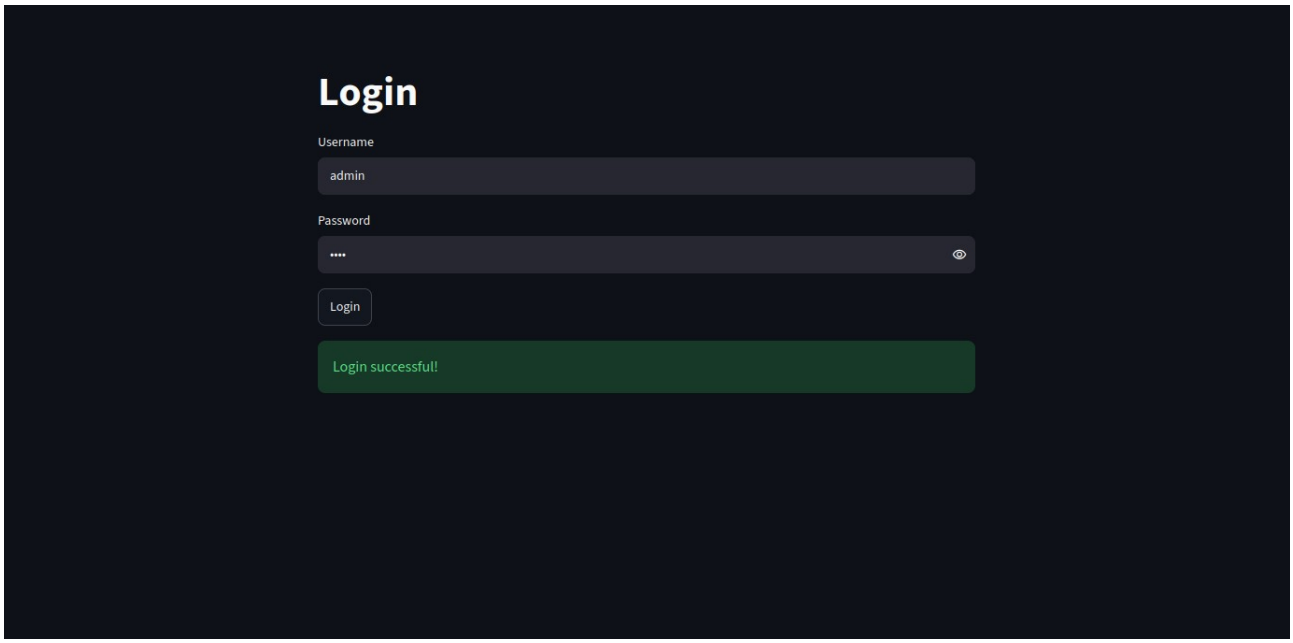

if st.button("Login"):

    if username == "admin" and password == "1234":

        st.success("Login successful!")
else:

     st.error("Invalid username or password")
```

OUTPUT:



17.Create student registration form using streamlit.

PROGRAM:

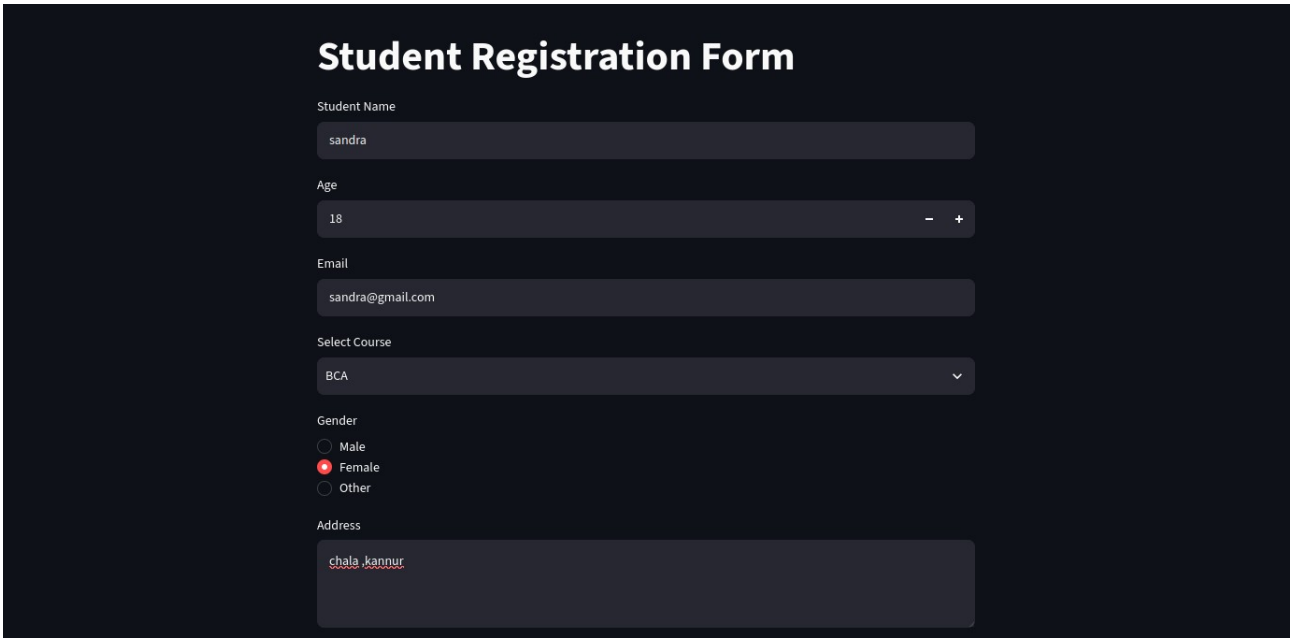
```
%%writefile resg-gui.py
import streamlit as st

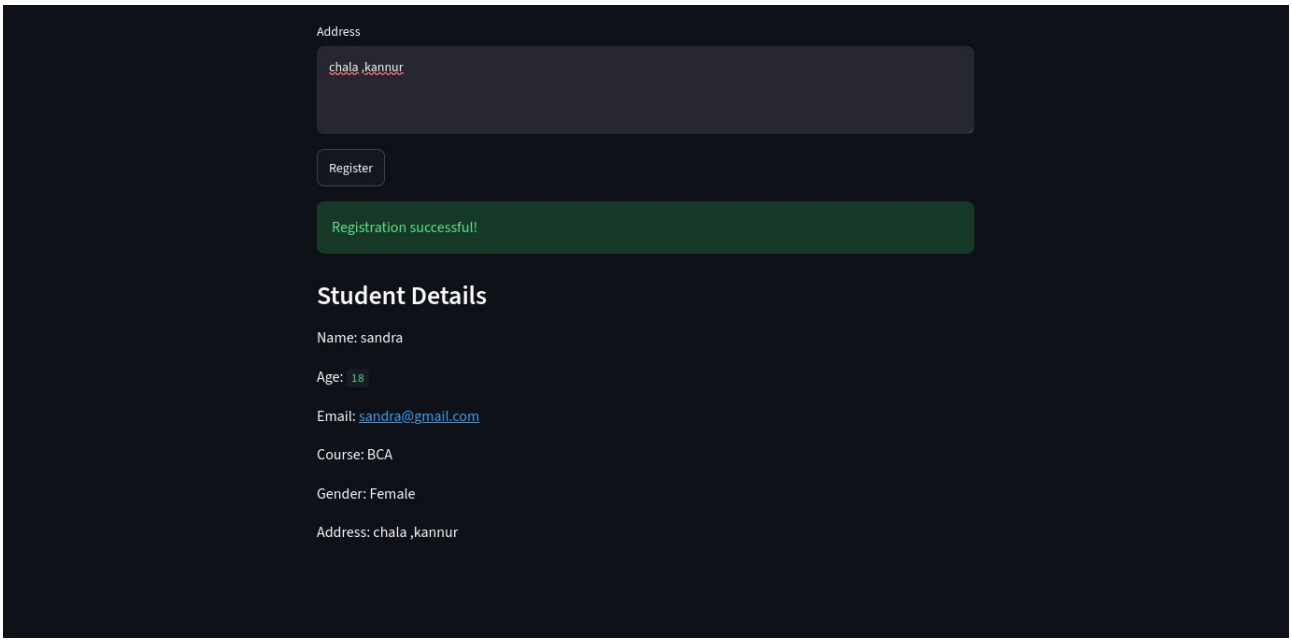
st.title("Student Registration Form")

name = st.text_input("Student Name")
age = st.number_input("Age", min_value=1, max_value=100)
email = st.text_input("Email")
course = st.selectbox(
"Select Course",
["BCA", "BSc Computer Science", "BCom", "BA"]
)
gender = st.radio("Gender", ["Male", "Female", "Other"])
address = st.text_area("Address")

if st.button("Register"):
    if name and email and address:
        st.success("Registration successful!")
        st.write("### Student Details")
        st.write("Name:", name)
        st.write("Age:", age)
        st.write("Email:", email)
        st.write("Course:", course)
        st.write("Gender:", gender)
        st.write("Address:", address)
else:
        st.warning("Please fill in all required fields.")
```

OUTPUT:


Student Registration Form
Student Name
sandra
Age
18
Email
sandra@gmail.com
Select Course
BCA
Gender
Male
Female
Other
Address
chala ,kannur


Address
chala ,kannur
Register
Registration successful!
Student Details
Name: sandra
Age: 18
Email: sandra@gmail.com
Course: BCA
Gender: Female
Address: chala ,kannur